

Workflow - QIIME 2 - Longitudinal Study

QIIME 2 Workflow

QIIME2 webpage: <https://qiime2.org>

```
# Import Ion Torrent single-end fastq sequences
# Sequences file are in the directory 'casava-18-single-end-demultiplexed'

qiime tools import \
  --type 'SampleData[SequencesWithQuality]' \
  --input-path casava-18-single-end-demultiplexed \
  --source-format CasavaOneEightSingleLanePerSampleDirFmt \
  --output-path demux-single-endL.qza &

# Check the quality plot
qiime demux summarize \
  --i-data demux-single-endL.qza \
  --o-visualization demux-single-endL.qzv &

# Denoise, dereplicate single-end sequences and remove chimeras
# Primer forward (trim-left) of 19 nucleotides
# Trunc-len value corresponds to the nucleotide truncation position and depends on the quality plot

qiime dada2 denoise-single \
  --i-demultiplexed-seqs demux-single-endL.qza \
  --p-trim-left 19 \
  --p-trunc-len --- \
  --p-n-threads 0 \
  --o-representative-sequences rep-seqs-dada.qza \
  --o-table table-dada.qza &

# check the previous outputs
qiime feature-table tabulate-seqs \
  --i-data rep-seqs-dada.qza \
  --o-visualization rep-seqs-dada.qzv &

qiime feature-table summarize \
  --i-table table-dada.qza \
  --m-sample-metadata-file metadata.tsv \
  --o-visualization table-dada.qzv &

# Look at the minimum number of samples sequences; we will use 9500 as sequences depth.
# Greengenes 13.8 at 99% of Operational Taxonomic Units (OTUs) identity and trimmed to V4 region

qiime feature-classifier classify-sklearn \
  --i-classifier ./classifier_GG99.qza \
  --i-reads rep-seqs-dada.qza \
  --p-n-jobs -1 \
  --o-classification taxonomy-clas-dada.qza &

# check the previous output
qiime metadata tabulate \
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--m-input-file taxonomy-clas-dada.qza \
--o-visualization taxonomy-clas-dada.qzv &

# Remove chloroplast sequences (Possible contaminants)
qiime taxa filter-table \
  --i-table table-dada.qza \
  --i-taxonomy taxonomy-clas-dada.qza \
  --p-exclude Chloroplast \
  --o-filtered-table tableGG-noC-dada.qza &

# check the previous output
qiime feature-table summarize \
  --i-table tableGG-noC-dada.qza \
  --o-visualization tableGG-noC-dada.qzv &

### Taxonomy (The last taxonomy script was performed using QIIME 1.9.1)
## The same following script were used for table L5 and L6

# L2
qiime taxa collapse
  --i-table tableGG-noC-dada.qza \
  --i-taxonomy taxonomy-clas-dada.qza \
  --p-level 2 \
  --o-collapsed-table tableCOLL-L2.qza &

qiime feature-table relative-frequency
  --i-table tableCOLL-L2.qza \
  --o-relative-frequency-table L2_relative.qza &

qiime tools export \
  L2_relative.qza \
  --output-dir exported-feature-table-L2 &

biom convert -i exported-feature-table-L2/feature-table.biom -o table.from_biom_w_taxonomy-L2.txt

### core-diversity (alpha and beta diversity)

qiime alignment mafft \
  --i-sequences rep-seqs-dada.qza \
  --o-alignment aligned-rep-seqs.qza &

qiime alignment mask \
  --i-alignment aligned-rep-seqs.qza \
  --o-masked-alignment masked-aligned-rep-seqs.qza &

qiime phylogeny fasttree \
  --i-alignment masked-aligned-rep-seqs.qza \
  --p-n-threads -1 \
  --o-tree unrooted-tree.qza &

qiime phylogeny midpoint-root \
  --i-tree unrooted-tree.qza \
  --o-rooted-tree rooted-tree.qza &

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# Sampling depth corresponds to the minimum samples sequences value
qiime diversity core-metrics-phylogenetic \
  --i-phylogeny rooted-tree.qza \
  --i-table tableGG-noC-dada.qza \
  --p-sampling-depth 9500 \
  --m-metadata-file metadata.tsv \
  --output-dir core-metrics &

### q2-longitudinal: a QIIME 2 plugin for longitudinal studies      (https://www.biorxiv.org/conte
## 'shannon_vector.qza', 'observed_otus_vector.qza', 'unweighted_unifrac_distance_matrix.qza' and

## Alpha diversity: differences within groups of samples
qiime longitudinal pairwise-differences \
  --m-metadata-file metadata_SM.tsv \
  --m-metadata-file shannon_vector.qza \
  --p-metric shannon \
  --p-group-column Treatment \
  --p-state-column Time \
  --p-state-1 T0 \
  --p-state-2 T5 \
  --p-individual-id-column Animal_Pair \
  --p-replicate-handling random \
  --o-visualization shannon-pairwise-differences.qzv &

qiime longitudinal pairwise-differences \
  --m-metadata-file metadata_SM.tsv \
  --m-metadata-file observed_otus_vector.qza \
  --p-metric observed_otus \
  --p-group-column Treatment \
  --p-state-column Time \
  --p-state-1 T0 \
  --p-state-2 T5 \
  --p-individual-id-column Animal_Pair \
  --p-replicate-handling random \
  --o-visualization obs_otus-pairwise-differences.qzv &

## Beta diversity: differences between groups of samples

qiime longitudinal pairwise-distances \
  --m-metadata-file metadata_SM.tsv \
  --i-distance-matrix unweighted_unifrac_distance_matrix.qza \
  --p-group-column Treatment \
  --p-state-column Time \
  --p-state-1 T0 \
  --p-state-2 T5 \
  --p-individual-id-column Animal_Pair \
  --p-replicate-handling random \
  --output-dir beta_div_distances_unwei/ &

qiime longitudinal pairwise-distances \

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--m-metadata-file metadata_SM.tsv \
--i-distance-matrix weighted_unifrac_distance_matrix.qza \
--p-group-column Treatment \
--p-state-column Time \
--p-state-1 T0 \
--p-state-2 T5 \
--p-individual-id-column Animal_Pair \
--p-replicate-handling random \
--output-dir beta_div_distances_wei &

## Beta diversity: differences within groups of samples

qiime tools export \
./ALPHA_BETA_NONEGS/core-metrics-NONEGS-SM/unweighted_unifrac_pcoa_results.qza \
--output-dir ./ALPHA_BETA_NONEGS/core-metrics-NONEGS-SM/PCOA_unwei/ &

qiime tools export \
./ALPHA_BETA_NONEGS/core-metrics-NONEGS-SM/weighted_unifrac_pcoa_results.qza \
--output-dir ./ALPHA_BETA_NONEGS/core-metrics-NONEGS-SM/PCOA_wei/ &

# the output (inside the folder PCOA_unwei or PCOA_wei) had to be .tsv and the header '#SampleID'

qiime longitudinal pairwise-differences \
--m-metadata-file metadata_SM.tsv \
--m-metadata-file PCOA_wei_pair.tsv \
--p-metric PC1 \
--p-group-column Treatment \
--p-state-column Time \
--p-state-1 T0 \
--p-state-2 T5 \
--p-individual-id-column Animal_Pair \
--p-replicate-handling random \
--o-visualization Wei-pairwise-differences-pcoa1-PAIRED.qzv &

qiime longitudinal pairwise-differences \
--m-metadata-file metadata_SM.tsv \
--m-metadata-file PCOA_wei_pair.tsv \
--p-metric PC2 \
--p-group-column Treatment \
--p-state-column Time \
--p-state-1 T0 \
--p-state-2 T5 \
--p-individual-id-column Animal_Pair \
--p-replicate-handling random \
--o-visualization Wei-pairwise-differences-pcoa2-PAIRED.qzv &

qiime longitudinal pairwise-differences \
--m-metadata-file metadata_SM.tsv \
--m-metadata-file PCOA_wei_pair.tsv \
--p-metric PC3 \
--p-group-column Treatment \
--p-state-column Time \
--p-state-1 T0 \
--p-state-2 T5 \

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--p-individual-id-column Animal_Pair \
--p-replicate-handling random \
--o-visualization Wei-pairwise-differences-pcoa3-PAIRED.qzv &

qiime longitudinal pairwise-differences \
--m-metadata-file metadata_SM.tsv \
--m-metadata-file PCOA_unwei_pair.tsv \
--p-metric PC1 \
--p-group-column Treatment \
--p-state-column Time \
--p-state-1 T0 \
--p-state-2 T5 \
--p-individual-id-column Animal_Pair \
--p-replicate-handling random \
--o-visualization unWei-pairwise-differences-pcoa1-PAIRED.qzv &

qiime longitudinal pairwise-differences \
--m-metadata-file metadata_SM.tsv \
--m-metadata-file PCOA_unwei_pair.tsv \
--p-metric PC2 \
--p-group-column Treatment \
--p-state-column Time \
--p-state-1 T0 \
--p-state-2 T5 \
--p-individual-id-column Animal_Pair \
--p-replicate-handling random \
--o-visualization unWei-pairwise-differences-pcoa2-PAIRED.qzv &

qiime longitudinal pairwise-differences \
--m-metadata-file metadata_SM.tsv \
--m-metadata-file PCOA_unwei_pair.tsv \
--p-metric PC3 \
--p-group-column Treatment \
--p-state-column Time \
--p-state-1 T0 \
--p-state-2 T5 \
--p-individual-id-column Animal_Pair \
--p-replicate-handling random \
--o-visualization unWei-pairwise-differences-pcoa3-PAIRED.qzv &

```

Taxonomic analysis - R workflow

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### For paired data
# install and load coin package
install.packages("coin")
library(coin)

# Load the taxonomy table including metadata information. Remember to divide files depending on the
a = L2+metadata_only_antibiotic

# Create an empty vector for p-values and the variable
outP=c()
outV=c()

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# For all taxonomic columns (from column 6 to the end) perform the pairwise.wilcox.test depending
for (i in c(6:dim(a)[2])){
  s=as.vector(as.matrix(a[,i]))
  m=as.vector(as.matrix(a[,5]))
  res=pairwise.wilcox.test(s, m, p.adj="none",exact=F, paired=T)
  outP = c(outP, res$p.value)
  outV = c(outV, rep(colnames(a)[i]))
}

outpvalue = data.frame(variable=outV, pvalues=outP)

View(outpvalue)

# Save the new file
write.csv2(outpvalue, 'my_output_table_L2+metadata_only_antibiotic.csv')

### For unpaired data we used the dunn test

# Install and load the dunn package
install.packages("dunn.test")
library(dunn.test)

The script are the same above: write 'dunn.test' instead of pairwise.wilcox.test

```